

ECE422 COMPUTER COMMUNICATIONS & NETWORKING — QUIZ PREP

Based on lecture slides: Ch.1 Introduction · Internet History · Protocols & Standards · Ch.2 Links, Encoding, Framing, Error Detection

Name: _____

Date: _____

Score: _____ / 100

PART I — IMPORTANT TOPICS (FROM THE SLIDES)

1. Layered model (Internet / TCP/IP)

- **TCP/IP has 4 layers** (per slides): **Application, Transport, Network, (Data) Link**. **Hosts implement all 4; routers only the bottom two** (Network + Link).
- **OSI has 7 layers**: Application, Presentation, Session, Transport, Network, Data Link, Physical.
- **Reasons for layering**: divide network operation into less complex elements; standard interfaces for plug-and-play / multi-vendor; modular development.
- A structured set of protocols = **communications architecture** or **protocol suite**. Protocols are independent, each does one task, grouped in a hierarchy.

Layer	Service	Functions
Application	Handles details of application programs	Everything application-specific
Transport	Controls delivery of data between hosts	Connection establishment/termination, error control, flow control
Network	Moves packets inside the network	Routing, addressing, switching, congestion control
Data Link	Reliable transfer of frames over a link	Synchronization, error control, flow control

- **Encapsulation**: each layer adds its header to the PDU from the layer above. **Overhead ratio** = header bytes / total bytes (state your assumption).

Connection-oriented	Connectionless (datagram)
Like the telephone system; call setup	Like the postal service; no call setup
Route set once for all packets	Every packet routed individually
Guaranteed delivery & sequence	No guarantees
Error/flow control by the network	Error/flow control by higher layers (end user)

- **Basic mechanisms**: Multiplexing (TDM, FDM, Statistical), Routing/Switching, Error Control (detection: parity, CRC; correction), Flow Control (window), Resource Allocation.
- **network** = connected communicating devices; **internet** = two or more networks that communicate; **the Internet** = the most notable one (since 1969). ISP = Internet Service Provider, NAP = Network Access Point.

2. Protocols & standards

- **Protocol** = rule(s); strict adherence. **Standard** = agreed-upon rules, established by authority, custom or general consent; developed by committees, forums, government/regulatory agencies.

- **Internet standard** = thoroughly tested specification. Starts as an **Internet Draft** (no official status, **six-month lifetime**) → may be published as an **RFC** (Request for Comment).
- **RFC maturity levels:** Proposed standard → Draft standard → Internet standard; plus Historic, Experimental, Informational.
- **RFC requirement levels:** Required, Recommended, Elective, Limited use, Not recommended.
- **Internet administration:** ISOC (Internet Society), IAB (Architecture Board), IETF (Engineering Task Force), IESG (Eng. Steering Group), IRTF (Research Task Force), IANA (Assigned Numbers Authority), ICANN, NIC (Network Information Center).

3. Internet history (dates from the slides)

Year	Event
1847 / 1876	Boolean algebra (Boole) / telephone (Bell) — phone exchanges became the Internet backbone, modems
1947	First working transistor, Bell Labs (Shockley, Bardeen, Brattain)
1957	USSR launches Sputnik → US forms ARPA within DoD
1962–68	Packet switching developed — military origin: no single outage point, packets take different routes, hard to eavesdrop
1969	ARPANET — 4 nodes: UCLA, SRI, UCSB, Utah; IMPs (Interface Message Processors). Oct 29, 1969: "LO" then crash (Kleinrock)
1971	E-mail invented; 15 nodes (23 hosts)
1972	Public ARPANET demo (40 machines); INWG created; Telnet spec
1973	TCP/IP invented (superseded NCP); 1973–76 Ethernet (Metcalfe & Boggs)
1982	TCP/IP established as the protocol suite for ARPANET
1984	Hosts > 1,000; DNS introduced; Cisco founded
1990	ARPANET ceases; Archie; The World = first commercial dial-up provider
1991	WWW released by CERN (Tim Berners-Lee); CIX; WAIS; Gopher; NSFNET T3; 2G cellular
1993 / 1994	MOSAIC GUI browser / commercialization begins
1996–99	Google ("Backrub"), "cloud computing" term (Compaq), Akamai, Napster (P2P)
2007 / 2008	iPhone / Android; IANA adds IPv6 records
2011 / 2012	Last IPv4 address assigned / IPv6 launch
2022	ChatGPT released

4. Links

- **Shannon–Hartley:** $C = B \cdot \log_2(1 + S/N)$ bps. $\text{dB} = 10 \cdot \log_{10}(S/N) \rightarrow 30 \text{ dB}$ means $S/N = 1000$.
Example (slide): $B = 3300 - 300 = 3000 \text{ Hz}$, $30 \text{ dB} \rightarrow C \approx 30 \text{ kbps}$.
- **Wavelength** = speed / frequency. On copper, speed $\approx 2/3 \cdot 3 \times 10^8 \text{ m/s}$. Slide: 300 Hz on copper $\rightarrow 2 \times 10^8 / 300 \approx 667 \text{ km}$.
- Media: copper (DSL, coax), optical fiber, free space (wireless). **Encoding** = placing binary data on a signal; **modulation** = modifying frequency, amplitude, phase.

5. Encoding

- **NRZ:** 1 = high, 0 = low. Problems: **baseline wander** (long runs change the receiver's average) and **clock recovery** (no transitions → lose sync).
- **NRZI:** transition to encode 1, stay to encode 0. Solves consecutive 1s, not consecutive 0s.
- **Manchester (slides' convention):** XOR of NRZ data and the clock. 0 = low → high, 1 = high → low (mid-bit). Problem: doubles transitions → **bit rate = half the baud rate** (50% efficient).
- **4B/5B:** every 4 data bits → 5-bit code; no more than **one leading 0** and **two trailing 0s** → never more than **three consecutive 0s**. Sent with NRZI. 80% efficient. Examples: 0000 → 11110, 0001 → 01001, 0010 → 10100, 1111 → 11101. 11111 = idle, 00000 = dead, 00100 = halt. Of 16 unused codes: 7 invalid, 6 control.

6. Framing

- Frames (blocks of data) are exchanged between nodes; the **adaptor** must recognize where a frame begins and ends.
- **Byte-oriented protocols:** BISYNC (IBM, late 1960s), DDCMP (DECNet).
- **BISYNC sentinel approach:** SYN (synchronize), SOH (start of header), STX (start of text), ETX (end of text), DLE (data link escape — character stuffing), CRC.

7. Error detection — overview

- Bit errors come from electrical interference and thermal noise. On error: (1) **notify sender** → **retransmit**, or (2) **correct** it at the receiver.
- Idea: add **redundant bits** (no new information, derived by an algorithm both sides know). Receiver recomputes r from m; match → no error.
- Sending two full copies = poor scheme (n redundant bits, errors can still go undetected). Good schemes: k redundant bits, $k \ll n$. Ethernet: up to 12,000 bits protected by a 32-bit CRC.
- Where used: **CRC** → HDLC, DDCMP, CSMA/CD, Token Ring · **2-D parity** → BISYNC · **Checksum** → IP (not at link level).

8. Single (1-D) parity

- Add one bit to a 7-bit code (8th bit). **Even parity:** total number of 1s even. **Odd parity:** total number of 1s odd.
- Detects any **odd** number of bit errors; misses **every even** number (each flip changes the count by ± 1 , so an even number of flips keeps even/odd the same).
- Number of k-bit error patterns in an n-bit word (n includes P) = $C(n,k)$.

9. Two-dimensional parity

- Parity bit for each byte (row) **plus** a parity byte for each bit position across all bytes (column).
- Catches **all 1-, 2-, 3-bit errors and most 4-bit errors**. Misses 4 errors on the corners of a rectangle. Can **locate and correct** a single-bit error (failing row × failing column).
- With odd parity and an odd×even block, the corner bit computed from the row vs. column differs — state which you used.

10. Internet checksum

- Treat data as **16-bit** integers; add with **ones-complement arithmetic** (carry out of the MSB is added back to the result); checksum = **ones complement** of the sum.

- Ones complement: $-x$ = invert every bit of x . Slide example (4-bit): $-5 = 1010$, $-3 = 1100$; $1010 + 1100 = 1\ 0110 \rightarrow 0110 + 1 = \mathbf{0111} = -8$.
- Receiver adds everything including the checksum; complement = 0 $\rightarrow$ OK.

11. CRC (Cyclic Redundancy Check)

- Bit string $\leftrightarrow$ polynomial: $110001 = x^5 + x^4 + 1$ (degree 5). A k -bit frame has max degree $k-1$.
- $M(x)$ = message, $C(x)$ = generator of degree r . Only **$C(x)$** must be known by both sides.
- **Steps:** append **r zeros** ($x^r M(x)$) $\rightarrow$ divide by $C(x)$ with **modulo-2 (XOR)** division $\rightarrow$ subtract (XOR) the remainder ($\leq r$ bits) $\rightarrow$ transmit. Receiver divides; **remainder $\neq 0 \rightarrow$ error**.
- Received = $P(x) + E(x)$. Error undetected only if $C(x)$ divides $E(x)$.
- **Detects:** all single-bit errors (if x^k and x^0 terms nonzero); all double-bit errors (if $C(x)$ has a factor with ≥ 3 terms); any odd number of errors (if $C(x)$ contains factor $(x+1)$); any burst shorter than k bits.
- Standards: CRC-8 = $x^8 + x^2 + x + 1$, CRC-10, CRC-12, CRC-16 = $x^{16} + x^{15} + x^2 + 1$, CRC-CCITT = $x^{16} + x^{12} + x^5 + 1$, CRC-32.
- Corrupt frames are discarded; reliability is recovered with **acknowledgements and timeouts**.

PART II — PRACTICE QUIZ

A. Encoding & Links

1. Represent **01101001** as unipolar NRZ and Manchester coding. [HW]

-
2. Represent **10110010** as NRZ, NRZI (assume the line starts low) and Manchester.

-
3. What are the two problems of NRZ? How do NRZI and Manchester address them, and which problem does NRZI still have?

-
4. A link sends 10 Mbps of data using Manchester encoding. What is its baud rate? What is the efficiency?

-
5. Using the 4B/5B table from the slides, encode **0000 0010 1111**. What is the efficiency of 4B/5B? What rule do the 5-bit codes follow?

-
6. Shannon: find the capacity for (a) $B = 3000$ Hz, $\text{SNR} = 30$ dB; (b) $B = 4000$ Hz, $\text{SNR} = 20$ dB.

-
7. Find the wavelength of (a) a 3300 Hz signal on copper, (b) a 2.4 GHz signal in free space.

B. Single (1-D) Parity

8. Add the 8th parity bit to each 7-bit code using (i) even and (ii) odd parity: **1011001**, **1110000**, **0000000**.

9. Using **odd** parity, the receiver gets **11010110** and **10011100**. Which one is in error? Could the other one still contain errors?

10. Which two-bit errors would not be detected for string **1001P**, if P is the **odd** parity bit? Indicate all the strings that contain those errors. [HW]

11. For **1101P** with **even** parity: how many 1-, 2-, 3- and 4-bit errors are possible, and which are detected?

C. Two-Dimensional Parity

12. Show the bi-dimensional parity of the codes **00010101**, **10110011**, **01000001** using odd parity. [HW]

13. Show the bi-dimensional parity of **11001010**, **01110001**, **10100110** using **even** parity.

14. The block below uses **even** 2-D parity (last column = row parity, last row = column parity). One bit was flipped. Find and correct it.

1	0	1	1		1
0	1	0	0		0
1	1	0	0		0

0	0	0	1		1

15. 2-D parity catches all 1-, 2-, 3-bit errors. Show a 4-bit error it cannot catch.

D. Checksum

16. In 4-bit ones-complement arithmetic, compute $-6 + (-2)$ (same style as the slide's $-5 + -3$).

17. Indicate the checksum field used for the transmission of the codes in Q12. [HW]

18. Compute the 16-bit Internet checksum of the words **1010 0011 1100 0101** and **0110 1101 1001 1010**.

E. CRC

19. Write the polynomial and degree for **110001** and **1011011**.

20. Message **1101**, generator $C(x) = x^3 + x + 1$. Find the CRC and the transmitted frame.

21. Message **10011010**, generator $C(x) = x^3 + x^2 + 1$. Find the transmitted frame.

22. With $C(x) = x^3 + x + 1$, the receiver gets **1101011**. Is there an error?

23. True / False:

- (a) Sender and receiver must both know $M(x)$.
- (b) All single-bit errors are detected if the x^k and x^0 terms of $C(x)$ are nonzero.
- (c) Any odd number of errors is detected if $C(x)$ contains the factor $(x+1)$.
- (d) The Internet checksum is used at the link level.
- (e) If $C(x)$ has degree r , the CRC has r bits.

F. Layers, Protocols, History

24. Assuming the TCP/IP layered model, if every layer encapsulates the PDU from the layer above and each header is 20B, indicate the overhead (as a ratio) for sending a) 1B and b) 100B of application data. [HW]

25. Indicate what would represent the layers of the TCP/IP model for an autonomous robot that imitates human walking. [HW]

26. Multiple choice.

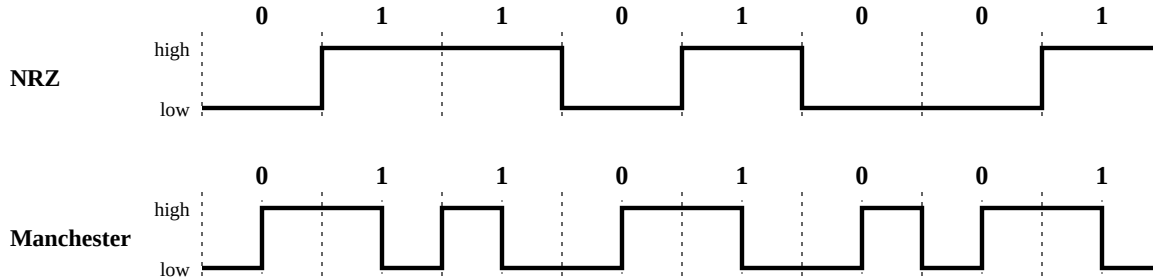
- (a) Number of layers in the TCP/IP suite (per slides): i) 4 ii) 5 iii) 7
- (b) Routers implement: i) all layers ii) the bottom two layers iii) only the link layer
- (c) An Internet Draft has a lifetime of: i) 1 month ii) 6 months iii) 1 year
- (d) 2-D parity is used in: i) IP ii) BISYNC iii) Token Ring
- (e) First ARPANET node: i) Stanford ii) MIT iii) UCLA
- (f) TCP/IP was invented in: i) 1969 ii) 1973 iii) 1991
- (g) The WWW was released by: i) CERN ii) ARPA iii) IETF
- (h) Manchester encodes a 1 as: i) low $\rightarrow$ high ii) high $\rightarrow$ low iii) no transition

27. Short answer: (a) protocol vs standard; (b) connection-oriented vs connectionless; (c) three reasons for layering; (d) service and functions of the data link layer; (e) BISYNC: SYN, SOH, STX, ETX, DLE.

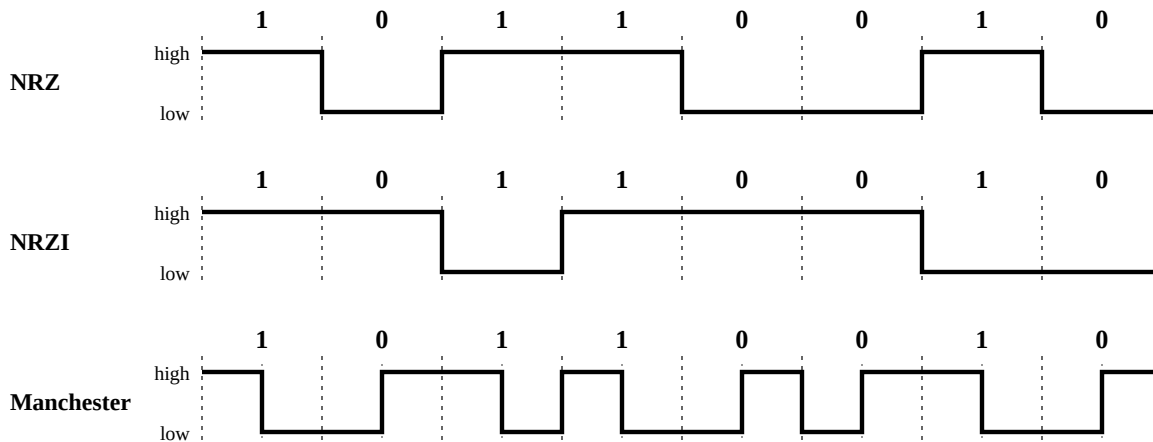
PART III — ANSWER KEY

Manchester convention used here = the slides' convention: 0 = low → high, 1 = high → low. (IEEE 802.3 is the opposite — always state which one you use.)

1. 01101001



2. 10110010



NRZI: flip on every 1, stay on 0 → levels H H L H H L L L.

3. **Baseline wander** (long runs of 0s/1s shift the receiver's average) and **clock recovery** (no transitions → sender/receiver lose sync). NRZI makes a transition for every 1 → fixes runs of 1s, but **runs of 0s still have no transitions**. Manchester has a transition in every bit → fixes both, but doubles the baud rate. (4B/5B + NRZI fixes runs of 0s.)

4. Two signal changes per bit → **20 Mbaud**. Bit rate = half the baud rate → efficiency **50%**.

5. 0000 → 11110, 0010 → 10100, 1111 → 11101 → **11110 10100 11101**. Efficiency $4/5 = 80\%$. Each code has at most one leading 0 and at most two trailing 0s, so the line never sees more than three 0s in a row.

6. (a) 30 dB → $S/N = 10^3 = 1000$ → $C = 3000 \cdot \log_2(1001) \approx 3000 \times 9.97 \approx 30 \text{ kbps}$.

(b) 20 dB → $S/N = 100$ → $C = 4000 \cdot \log_2(101) \approx 4000 \times 6.66 \approx 26.6 \text{ kbps}$.

7. (a) $\lambda = (2/3 \cdot 3 \times 10^8) / 3300 = 2 \times 10^8 / 3300 \approx 60.6 \text{ km}$. (b) $\lambda = 3 \times 10^8 / 2.4 \times 10^9 = 0.125 \text{ m}$ (12.5 cm).

8.

7-bit code	# of 1s	Even parity → byte	Odd parity → byte
1011001	4	P=0 → 10110010	P=1 → 10110011
1110000	3	P=1 → 11100001	P=0 → 11100000
0000000	0	P=0 → 00000000	P=1 → 00000001

9. 11010110 has 5 ones (odd) → passes. 10011100 has 4 ones (even) → **error**. The first one can still be wrong: an even number of flipped bits (2, 4, ...) is invisible to parity.

10. 1001 has two 1s → P = 1 → codeword **10011**. A single parity bit misses every 2-bit error → C(5,2) = **10** undetected errors:

Flipped	1,2	1,3	1,4	1,5	2,3	2,4	2,5	3,4	3,5	4,5
String	01011	00111	00001	00010	11111	11001	11010	10101	10110	10000

All have an odd number of 1s, so all pass the odd-parity check.

11. 1101 has three 1s → P = 1 → codeword 11011 (four 1s). 1-bit: C(5,1) = 5, **all detected**. 2-bit: C(5,2) = 10, **none detected**. 3-bit: C(5,3) = 10, **all detected**. 4-bit: C(5,4) = 5, **none detected**. Rule: odd number of flips → detected; even → missed.

12. Odd parity:

	Data	# of 1s	P
R1	0 0 0 1 0 1 0 1	3	0
R2	1 0 1 1 0 0 1 1	5	0
R3	0 1 0 0 0 0 0 1	2	1
Col P	0 0 0 1 1 0 0 0		1*

*Corner: odd parity of the Col P row (two 1s) → 1; odd parity of the P column (one 1) → 0. They disagree with odd parity because there are 3 rows and 8 columns — state the convention you used.

13. Even parity:

	Data	# of 1s	P
R1	1 1 0 0 1 0 1 0	4	0
R2	0 1 1 1 0 0 0 1	4	0
R3	1 0 1 0 0 1 1 0	4	0
Col P	0 0 0 1 1 1 0 1		0

14. Check each row: R1 1011|1 → four 1s ✓; R2 0100|0 → **one 1 ✗**; R3 1100|0 → two 1s ✓. Check each column: c1 1,0,1|0 ✓; c2 0,1,1|0 ✓; c3 1,0,0|0 → **one 1 ✗**; c4 1,0,0|1 ✓.

Error at **row 2, column 3** → correct R2 to **0110**. 2-D parity can correct a single-bit error.

15. Flip 4 bits on the corners of a rectangle, e.g. (R1,c1), (R1,c2), (R2,c1), (R2,c2). Each affected row and each affected column has **two** flips → every parity still checks → undetected.

16. +6 = 0110 → -6 = 1001; +2 = 0010 → -2 = 1101.

1001 + 1101 = 1 0110 → carry out, add it back: 0110 + 1 = **0111** = ones complement of 1000 = **-8 ✓**.

17.

```
00010101    (21)
+ 10110011    (179)
+ 01000001    (65)
= 1 00001001 (265) → wrap carry: 00001001 + 1 = 00001010
Complement → CHECKSUM = 11110101 (245)
Receiver: 00001010 + 11110101 = 11111111 → complement = 00000000 → OK
```

(8-bit words, as in the HW. The real Internet checksum uses 16-bit words.)

18.

```
1010 0011 1100 0101    (A3C5)
+ 0110 1101 1001 1010    (6D9A)
= 1 0001 0001 0101 1111 → wrap carry: 0001 0001 0101 1111 + 1 = 0001 0001 0110 0000
Complement → CHECKSUM = 1110 1110 1001 1111    (EE9F)
```

19. $110001 = x^5 + x^4 + 1$, degree 5. $1011011 = x^6 + x^4 + x^3 + x + 1$, degree 6.

20. $C(x) = 1011$, degree $r = 3 \rightarrow$ append 3 zeros: **1101000**.

```
1101000
1011      XOR → 0110 → drop the 0, bring down 0 → 1100
 1100
1011      XOR → 0111 → bring down 0 → 1110
 1110
1011      XOR → 0101 → bring down 0 → 1010
 1010
1011      XOR → 0001 → remainder = 001
```

CRC = **001** → transmitted frame **1101001**. (Check: $1101001 \div 1011$ leaves remainder 000.)

21. $C(x) = 1101$, $r = 3 \rightarrow$ **10011010000**.

```
1001 ⊗ 1101 = 0100 → 100, bring 1 → 1001
1001 ⊗ 1101 = 0100 → 100, bring 0 → 1000
1000 ⊗ 1101 = 0101 → 101, bring 1 → 1011
1011 ⊗ 1101 = 0110 → 110, bring 0 → 1100
1100 ⊗ 1101 = 0001 → 001, bring 0 → 0010 (starts with 0: no XOR)
                      010, bring 0 → 0100 (starts with 0: no XOR)
                      100, bring 0 → 1000
1000 ⊗ 1101 = 0101 → remainder = 101
```

Transmitted frame = **10011010 101**.

22.

```
1101 ⊗ 1011 = 0110 → 110, bring 0 → 1100
1100 ⊗ 1011 = 0111 → 111, bring 1 → 1111
1111 ⊗ 1011 = 0100 → 100, bring 1 → 1001
1001 ⊗ 1011 = 0010 → remainder = 010 ≠ 0
```

Remainder is not zero → **error detected**. (The correct frame was 1101001; bit 6 flipped.)

23. (a) **F** — only C(x) must be known. (b) **T**. (c) **T**. (d) **F** — the checksum is not used at the link level (it is used in IP). (e) **T**.

24. The slides' TCP/IP model has 4 layers. Transport, Network and Link each encapsulate the PDU from above → $3 \times 20 = 60\text{B}$ of headers.

a) 1B: $60 / (60 + 1) = 60/61 \approx 0.98$ b) 100B: $60 / 160 = 0.375$

If you also count an application header ($4 \times 20 = 80\text{B}$): $80/81 \approx 0.99$ and $80/180 \approx 0.44$. **State your assumption.**

25. Example with 4 layers:

Layer	Walking robot
Application	Goal / intent: "walk to the door", path planning
Transport	Gait control: splits the goal into steps, sequencing, confirms each step completed (feedback, error control)
Network	Chooses the route and foot placement across the terrain; coordinates which leg/joints move (routing, addressing)
Link	Joint-level controllers and signals to each motor, sensor frames, synchronization; the physical motors, wires and legs

HW asks for two versions: yours first, then an LLM's response.

26. (a) i 4 (b) ii bottom two (c) ii 6 months (d) ii BISYNC (e) iii UCLA (f) ii 1973 (g) i CERN (h) ii high → low

27. (a) Protocol = rules for communication, followed strictly; standard = agreed-upon rules set by authority, custom or consensus (committees, forums, agencies).

(b) Connection-oriented: call setup, one route for all packets, guaranteed delivery and order, error/flow control by the network (telephone). Connectionless: no setup, each packet routed separately, no guarantees, error/flow control by the end hosts (postal service).

(c) Break network operation into simpler parts; standard interfaces for plug-and-play and multi-vendor; modular development.

(d) Service: reliable transfer of frames over a link. Functions: synchronization, error control, flow control.

(e) SYN = synchronize (frame start), SOH = start of header, STX = start of text, ETX = end of text, DLE = data link escape (character stuffing when ETX appears in the data).

Last-minute checklist: Manchester per slides ($0 = L \rightarrow H$, $1 = H \rightarrow L$) · NRZI flips on 1 · $4\text{B}/5\text{B} = 80\%$, Manchester = 50% · parity misses even numbers of errors, count with C(n,k), n includes P · 2-D parity: all 1/2/3-bit errors, corrects 1 · checksum: add, wrap carry, complement · CRC: append r zeros, XOR divide, remainder = CRC, receiver remainder 0 = OK · TCP/IP = 4 layers, routers = bottom two · 1969 ARPANET, 1973 TCP/IP, 1984 DNS, 1991 WWW.